

NOIDA INSTITUTE OF ENGINEERING & TECHNOLOGY GREATER NOIDA



Course & Branch Project Management (AOE0761)

Unit 1: Introduction of Project

By

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Department of MBA
NIET, GREATER NOIDA**

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Qualification: M.Com

Specialization: Finance and Accounting

Teaching Area: Finance

Evaluation Scheme

MBA
EVALUATION SCHEME
SEMESTER-IV

No	Code	Subject	Periods			Evaluation Scheme				End Semester		Total	Credit
			L	T	P	CT	TA	PS	Total	TE	PE		
1	AMBA0401	Project Management	3	0	0	30	20	0	50	100	0	150	3
2		Specialization Group -1 Elective -4	3	1	0	30	20	0	50	100	0	150	4
3		Specialization Group -1 Elective -5	3	1	0	30	20	0	50	100	0	150	4
4		Specialization Group -1 Elective -6	3	1	0	30	20	0	50	100	0	150	4
5		Specialization Group -2 Elective- 3	3	1	0	30	20	0	50	100	0	150	4
6		Specialization Group -2 Elective- 4	3	1	0	30	20	0	50	100	0	150	4
7	AMBA0459	Research Project Report*	0	0	6				100		100	200	3
GRAND TOTAL												1100	26

* Satisfactory completion of minimum 1 'Research Publication' in a listed Journal is mandatory for award of degree.

Syllabus

UNIT-I	Introduction of Project	06 Hours
Projects – Definition and Objectives – Project Management Vs General Management – Roles and Responsibilities of Project Manager – Selection of Project Manager – Selection of Projects – Understanding Project Life Cycle. Project Team and Scope of Project Management: Characteristics of a Project Team & Project Leader, Project Organization, and Importance of Project Management. Case Studies		
UNIT-II	Project Identification & Selection	08 Hours
Project Identification & Selection: Identification, Generation of ideas, Approaches to Project Screening and Selection, Project Rating Index. Market & Demand Analysis Techniques: Survey & Trend Projection Methods. Project Risk Management: Concepts and Types of Project Risks, Risk Identification, Risks Analysis, Risks Mitigation Strategies. Case Studies		
UNIT-III	Budgeting the Project	08 Hours
Fundamental components of Project Cost, Types of Costs: Direct, Indirect, Recurring, Non-Recurring, Fixed, Variable, Normal, Expedite costs Methods of budgeting – Project cost estimation – Improving cost estimates – Budget uncertainty and risk management – Scheduling the project – Gantt chart – Resource allocation and loading – Social Cost Benefit Analysis (SCBA) of Project: Concept & significance of SCBA, Approaches to SCBA. Case Studies		
UNIT-IV	Project Scheduling and Network Analysis	08 Hours
Steps in Project Scheduling and Network design, Gantt Chart, Work Breakdown Structure (WBS) & Responsibility Assignment Matrix. Project Network Design: Identifying the Nodes and Activities, Activity on Arrow (AoA) and Activities on Node (AoN) methods, Introduction to PERT and CPM, Crashing in Projects. Case Studies		
UNIT-V	Monitoring and controlling the project	6 Hours
Monitoring the project – Control cycle – Project control – Designing the control system – Evaluation of project: Milestone Analysis and Tracking Gantt chart. Earned Value Analysis (EVA): Planned Value (PV), Earned Value (EV), Cost Variance (CV), Schedule Variance (SV), Cost performance Index (CPI), Schedule performance Index (SPI) – Project auditing – Project termination: Types of Terminations, Project Termination Process. Case Studies		

Branch wise Application

- Handling projects.
- Evaluating & Monitoring Project Process
- Taking necessary and corrective action.

Course Objectives

- To empower the students to get insights into basic concepts of project management.
- To create awareness of the roles and responsibilities of the project manager.
- To build confidence among the students to take up any kind of project.
- To sharpen the planning, scheduling, and controlling skills of the students concerning individual projects.
- To understand the perspectives in which optimum decisions are to be taken in case of risks with planned activities in the project.

Course Outcomes

Course outcome: At the end of course, the student will be able to:		
CO 1	Understand the basic concepts and characteristics of Project and Project manager, management	Understanding (K2)
CO 2	Understand the roles and responsibilities along with tools & techniques used in Project management	Evaluating (K5)
CO 3	Develop confident to take up any kind of projects	Evaluating (K5)
CO 4	Students will understand the scheduling and monitoring process in Project. They will be able to apply PERT and CPM method for project scheduling	Applying (K3)
CO 5	Students will understand the perspectives in which optimum decisions are to be taken in case of risks with planned activities in project	Creating (K6)

Program Outcome(s)

1. Apply knowledge of management theories and practices to solve business problems.
2. Foster analytical and critical thinking abilities for data-based decision-making.
3. Ability to develop value-based leadership ability.
4. Ability to understand, analyze, and communicate global, economic, legal, and ethical aspects of business.
5. Ability to lead themselves and others in the achievement of organizational goals, contributing effectively to a team environment.

COs and POs mapping

S No	CO/PO	PO1	PO2	PO3	PO4	PO5
1	CO1		3		2	
2	CO2			3		
3	CO3		3		1	
4	CO4			1		3
5	CO 5	1	2			

*1=High, *2=Medium, *3=Low



Program Educational Objectives

- PEO 1:** Graduates of the Management program will have conceptual knowledge, to adapt to the rapidly changing environment, learn new skills and demonstrate application of management principles in a professional work setting.
- PEO 2:** Graduates will apply appropriate tools for decision making required for solving complex managerial problems in the local or global context.
- PEO 3:** Graduates of the Management program will exhibit integrity, social responsibility and teamwork.
- PEO 4:** Graduates will exhibit ethics, communication skills, leadership qualities and entrepreneurial mindset using creativity and innovation.



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Result Analysis



End Semester Question Paper Template

Printed page:

Subject Code:

Roll No:

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NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY, GREATER NOIDA

(An Autonomous Institute Affiliated to AKTU, Lucknow)

B.Tech/B.Voc./MBA/MCA/M.Tech (Integrated)

(SEM:.....THEORY EXAMINATION(2020-2021))

Subject

Time: 3Hours Max. Marks:100

General Instructions:

- All questions are compulsory. Answers should be brief and to the point.
- This Question paper consists ofpages & ...8.....questions.
- It comprises of three Sections, A, B, and C. You are to attempt all the sections.
- **Section A** - Question No- 1 is objective type questions carrying 1 mark each, Question No- 2 is very short answer type carrying 2 mark each. You are expected to answer them as directed.
- **Section B** - Question No-3 is Long answer type -I questions with external choice carrying 6 marks each. You need to attempt any five out of seven questions given.
- **Section C** - Question No. 4-8 are Long answer type -II (within unit choice) questions carrying 10marks each. You need to attempt any one part a or b.
- Students are instructed to cross the blank sheets before handing over the answer sheet to the invigilator.
- No sheet should be left blank. Any written material after a blank sheet will not be evaluated/checked.

End Semester Question Paper Template

<u>SECTION – B</u>			CO
3.	Answer any <u>five</u> of the following-		[5×6=30]
3-a.	Question-	(6)	
3-b.	Question-	(6)	
3-c.	Question-	(6)	
3-d.	Question-	(6)	
3-e.	Question-	(6)	
3-f.	Question-	(6)	
3-g.	Question-	(6)	
<u>SECTION – C</u>			CO
4	Answer any <u>one</u> of the following-		[5×10=50]
4-a.	Question-	(10)	
4-b.	Question-	(10)	
5.	Answer any <u>one</u> of the following-		
5-a.	Question-	(10)	
5-b.	Question-	(10)	
6.	Answer any <u>one</u> of the following-		
6-a.	Question-	(10)	
6-b.	Question-	(10)	
7.	Answer any <u>one</u> of the following-		
7-a.	Question-	(10)	
7-b.	Question-	(10)	
8.	Answer any <u>one</u> of the following-		
8-a.	Question-	(10)	
8-b.	Question-	(10)	

Perquisite/Recap

- Understanding of general management



Brief Introduction about the Subject with Videos

- Projects are all around us. Virtually every organization runs projects, either formally or informally. Across settings, planning principles and execution methodologies can offer ways in which projects can be run more effectively and efficiently. Project management provides organizations (and individuals) with the language and the frameworks for scoping projects, sequencing activities, utilizing resources, and minimizing risks.
- <https://www.youtube.com/watch?v=uEREF70j-yI>
- https://www.youtube.com/watch?v=ewhTgzIQG84&list=PLLy_2iUCG87CBuNhvti0h6W54ZmqrSDMJ&index=5
- https://www.youtube.com/watch?v=fDh0K3SYT6A&list=PLLy_2iUCG87CBuNhvti0h6W54ZmqrSDMJ&index=21

Unit Content

- Projects – Definition and Objectives – Project Management Vs General Management – Roles and Responsibilities of Project Manager – Selection of Project Manager – Selection of Projects – Understanding Project Life Cycle. Project Team and Scope of Project Management: Characteristics of a Project Team & Project Leader, Project Organization, and Importance of Project Management

Objective of Unit

1. Understand the meaning of Project and Criteria of Good Project in Project Management.
2. Identify the various Characteristics of project.

Introduction

- A project is a temporary endeavour that produces a unique product, service, or result.
- Definitive start and finish dates.
- Project management is the application of tools and techniques to organize the project activities to successfully meet the project goals.



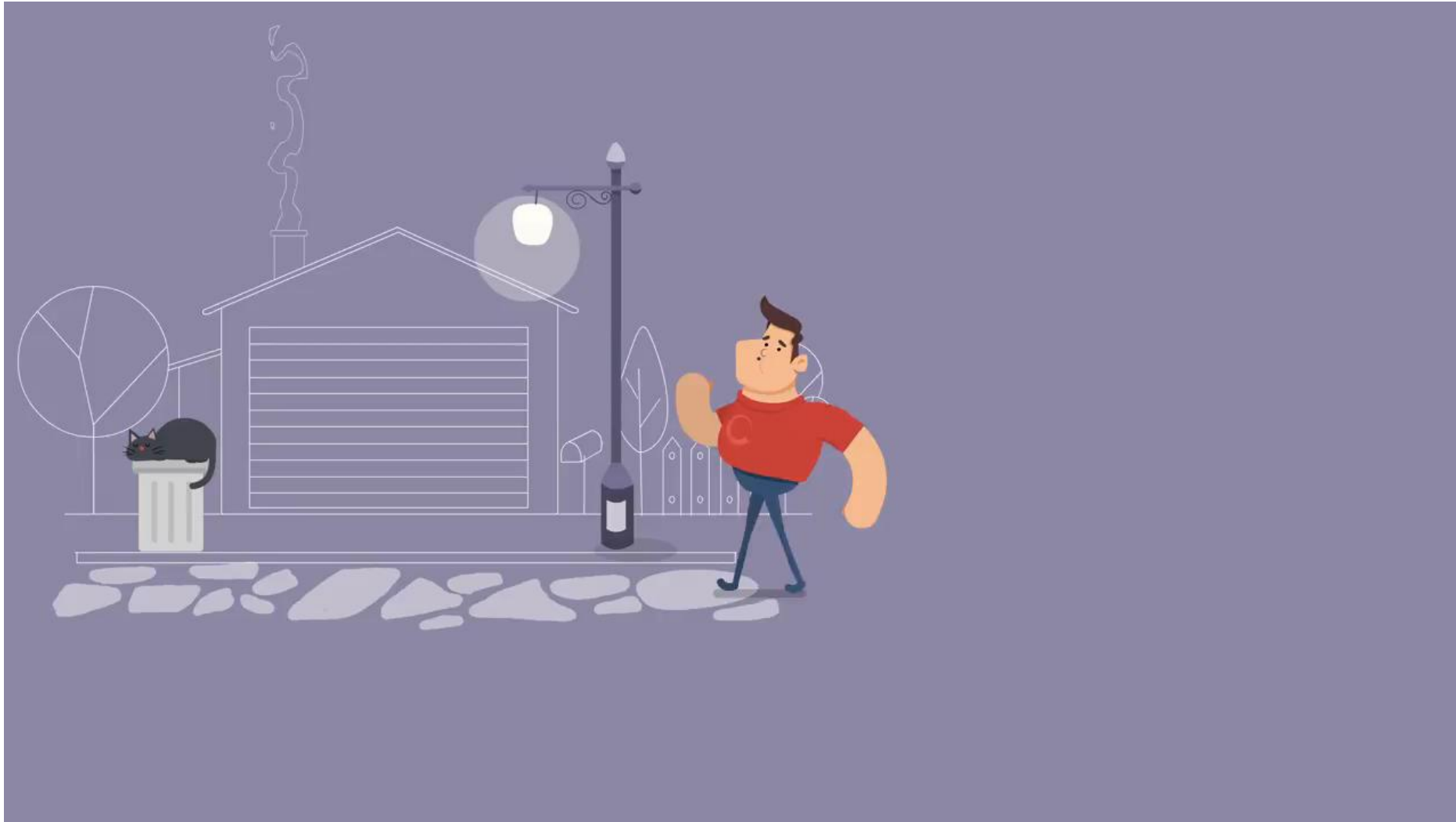
Project

- A project is defined as an effort to create or modify a specific product or service.
- Projects are temporary work efforts with a clear beginning and end.
- Projects can be completely contained within a specific unit or department or include other organizations and vendors.



More definitions of project

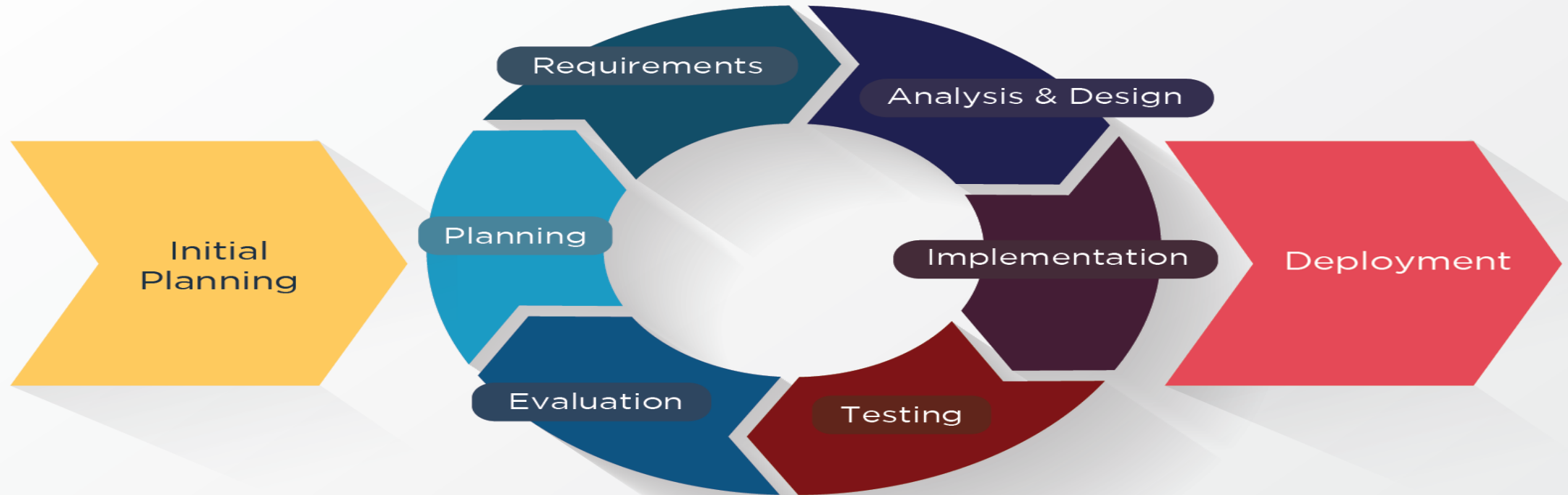
- **Project Management Institute** - 'A project is a temporary endeavor undertaken to create a unique product, service or result.'
- **Association for Project Management (APM)** - 'An endeavor in which human material and financial resources are organized in a novel way to deliver a unique scope of work of given specification often within constraints of cost and time to achieve beneficial changes defined by quantitative and qualitative objectives.'



More about project management

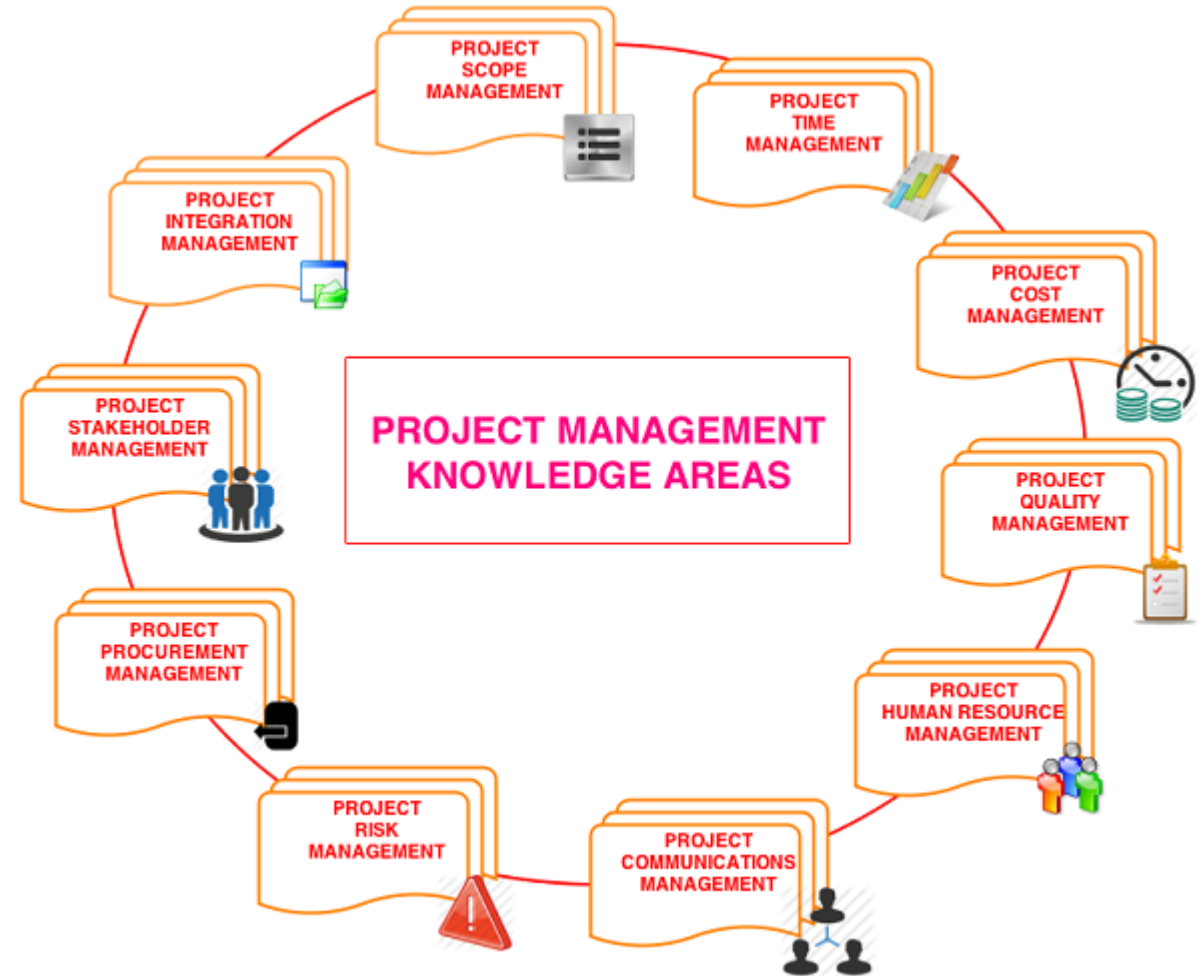


Project Management



Key Areas Of Project Management

- Scope Management
- Issue Management
- Cost Management
- Quality Management
- Communications Management
- Risk Management
- Change Control Management



Features/Characteristics/Components of Project

- **Scope**
- **Time**
- **Budget**
- **Schedule**
- **Resources**
- **Procurement**
- **Risk**
- **Stakeholders**

Project Stakeholders

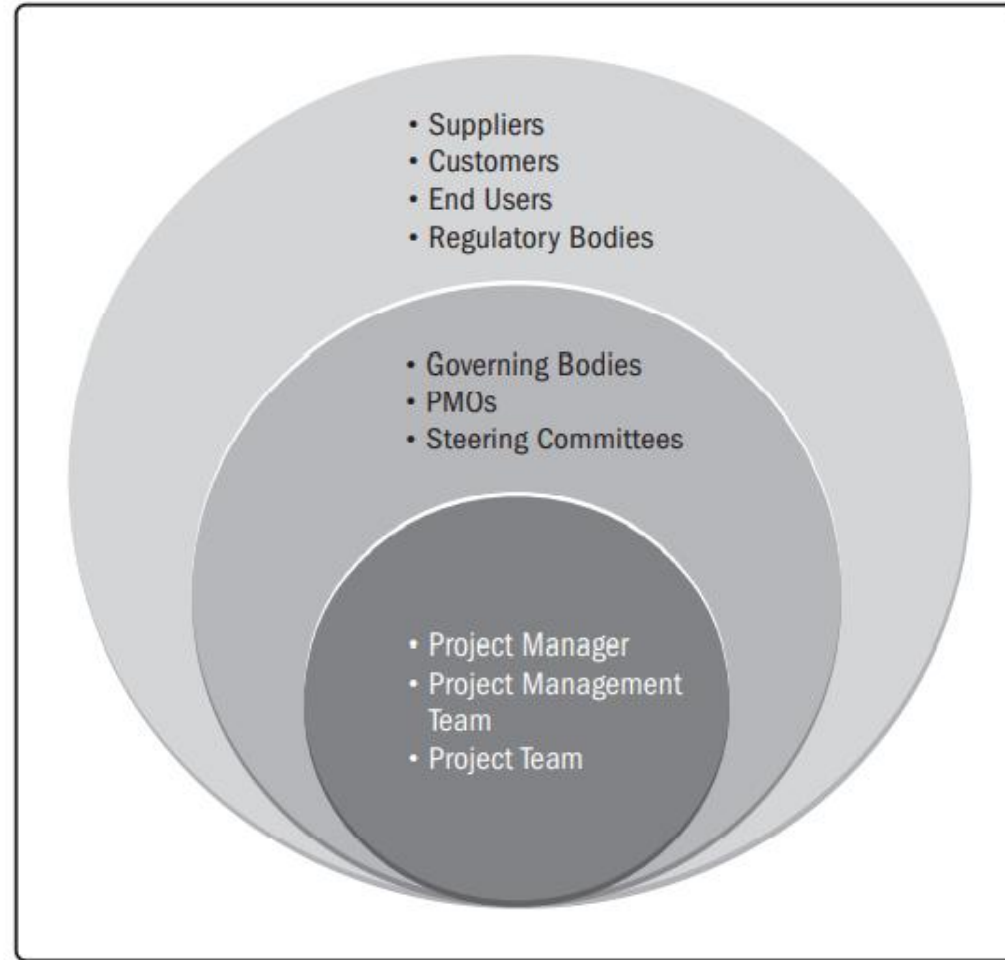


Figure 2-2. Examples of Project Stakeholders

Types of project

- **Engineering projects** encompass civil, electrical, and mechanical engineering, and the final deliverables are physical objects,
- Examples: a building, reservoir, bridge, refinery, or pre-production sample. Specialist companies or consortia invariably undertake these types of projects.
- **Management projects** include things like restructuring the organization, preparing for an exhibition, developing an IT system, launching a new marketing campaign, moving offices, or indeed anything where the objective is to produce a result that is not identifiable as a physical item

Basic Terminologies

- **Portfolio:** Projects, programs, subsidiary portfolios, and operations managed as a group to achieve strategic objectives.
- **Product:** An artifact that is produced, is quantifiable, and can be either an end item in itself or a component item.
- **Program:** Related projects, subsidiary programs, and program activities that are managed in a coordinated manner to obtain benefits not available from managing them individually.

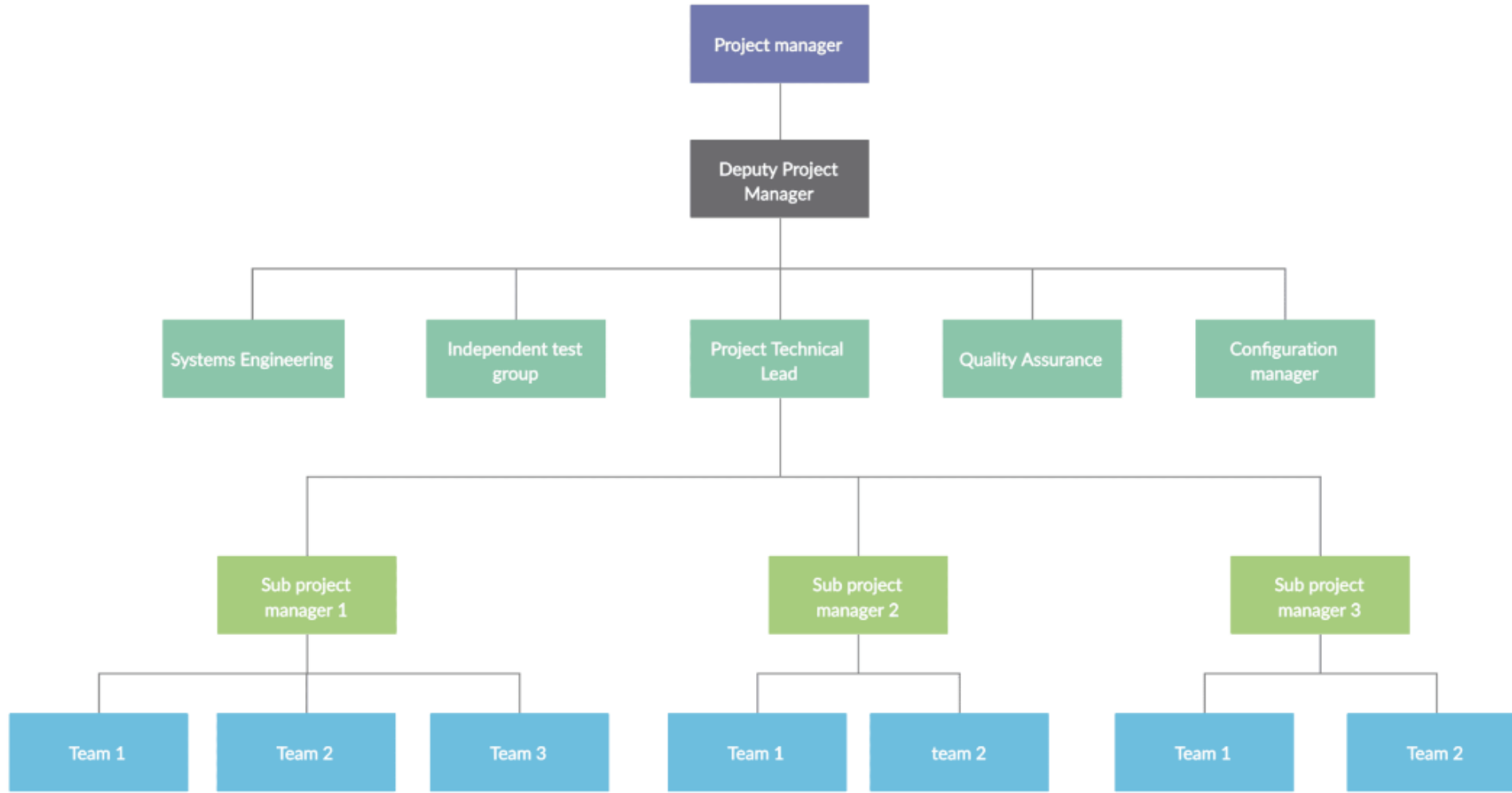
Basic Terminologies

- **Project**: A temporary endeavor undertaken to create a unique product, service, or result. The temporary nature of projects indicates a beginning and an end to the project work or a phase of the project work. Projects can stand alone or be part of a program or portfolio.
- **Project manager**: The person assigned by the performing organization to lead the project team that is responsible for achieving the project objectives. Project managers perform a variety of functions, such as facilitating the project team work to achieve the outcomes and managing the processes to deliver intended outcomes.

Basic Terminologies

- **Project management:** The application of knowledge, skills, tools, and techniques to project activities to meet project requirements. Project management refers to guiding the project work to deliver the intended outcomes.
- **Value:** The worth, importance, or usefulness of something. Different stakeholders perceive value in different ways.
 - Customers can define value as the ability to use specific features or functions of a product.
 - Organizations can focus on business value as determined with financial metrics, such as the benefits less the cost of achieving those benefits.
 - Societal value can include the contribution to groups of people, communities, or the environment.

Hierarchy



Sample System for value delivery

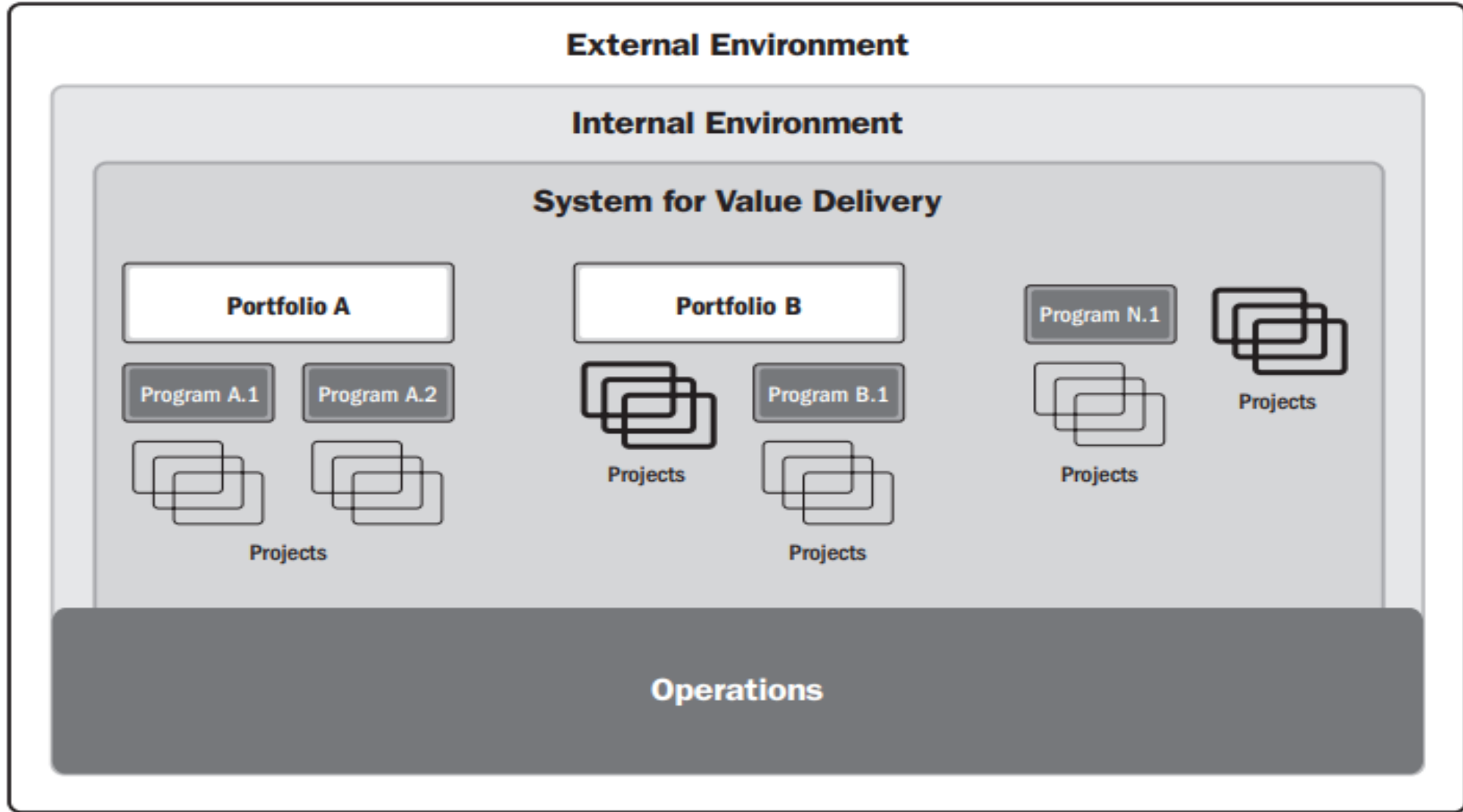


Figure 2-2. Components of a Sample System for Value Delivery

Information Flow

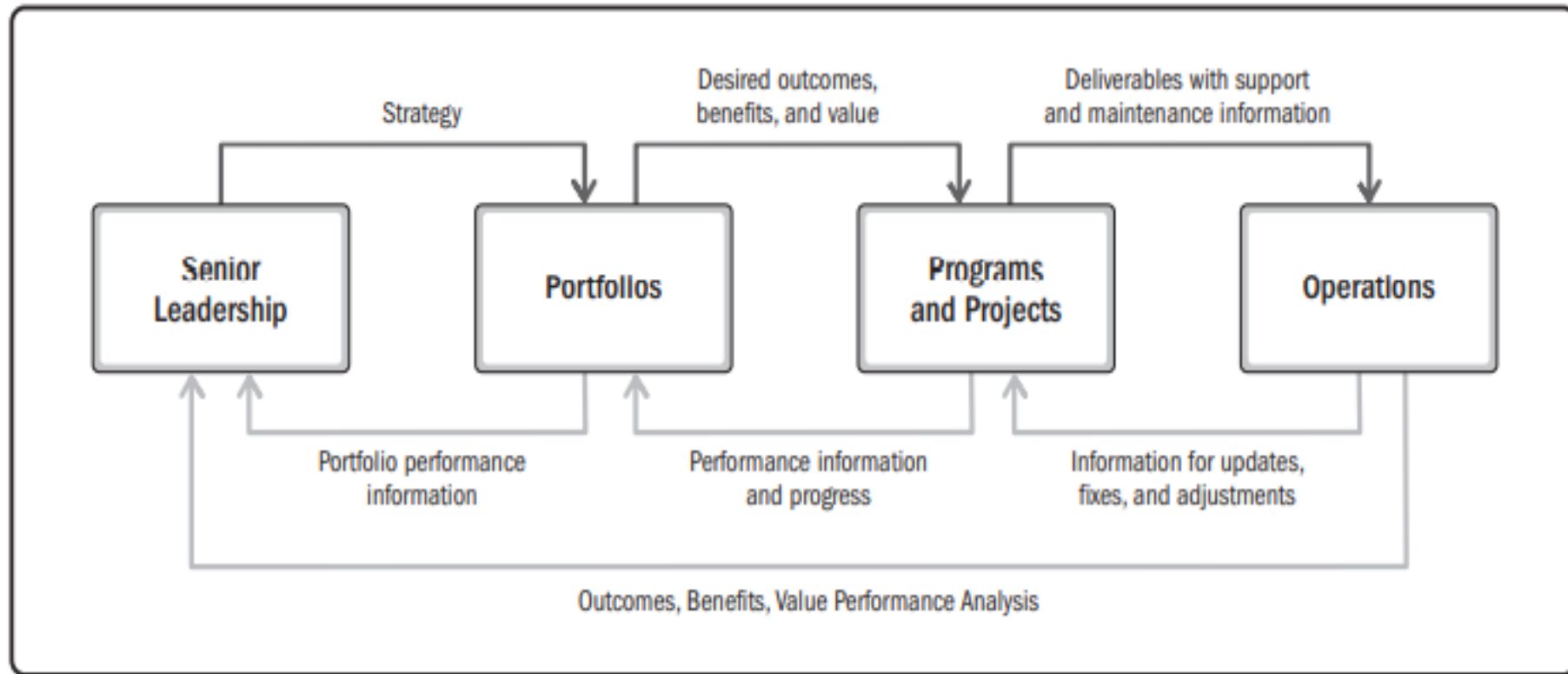


Figure 2-3. Example of Information Flow

- Define Project Management.
- Identify the various Characteristics of project.
- Discuss the types of Project.
- Discuss the project team.
- Explain the scope of Project Management.
- Who leads the team of project- a project manager or a leader?
- Is the idea a project?
- List the various types of projects.
- Define a project.

- The Meaning of Project Involved in Project Management.
- Project Team
- Project Manager
- The Various Characteristics of Project.
- The Formulation of Types of Project.
- The Project Team.
- The Scope of Project Management.

Project Success

- **Set Expectations:** Define the Project
- **Get Organized:** Create the Project Plan
- **Start at the Top:** Gain Executive Support
- **The Best Teams Deliver the Best Results:** Project Team Management
- **Prepare for the Inevitable:** Risk Management

Project Success

- **Stick to the Plan:** Scope Management
- **Deliver on the Plan:** Quality Assurance
- **Spread the News:** Communicate Early and Often
- **Gain User Acceptance:** Change Management
- **Quantify Results:** Measure Project Success

Project can fail due to:

- Lack of resource planning
- Unclear Goals and Objectives
- Lack of project visibility
- Communication gaps
- Scope creep
- Unrealistic expectations

Categories of Project

1.	Administrative	Installing a new accounting system
2.	Construction	Building or road
3.	Computer Software Development	New computer program
4.	Design of Plans	Architectural or engineering plans
5.	Equipment or System Installation	Telephone system or IT system
6.	Event or Relocation	Olympiads or a move into a new building
7.	Maintenance of Process Industries	Petro-chemical plant or electric generating station
8.	New Product Development	New drug or aerospace/defense product
9.	Research	Feasibility study or investigating a chemical

Project vs. Regular Operations

- **Project**

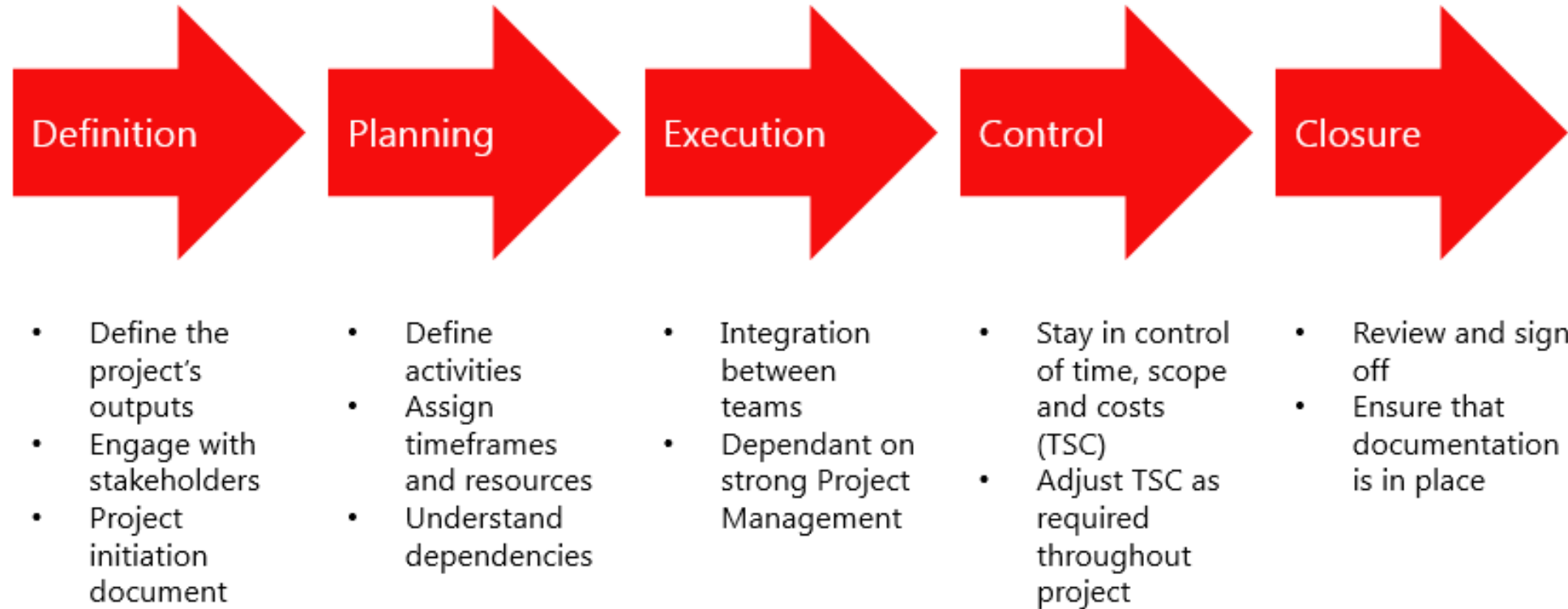
- temporary - they have a definitive beginning and end
- unique - they are a new undertaking, unexplored ground

- **Regular Operations**

- repetitive - they have expected inputs and outputs every time
- ongoing - they have a continuous cycle of execution

- State the difference between project and regular operations.
- Illustrate the factors that lead to failure of the project.

Project Life Cycle



Step 1: Definition

- Project manager must prove that the project has value and is feasible.
- A project charter, which is a document that conveys what the project is going to deliver, is created. A project brief serves a similar purpose.
- This project management stage culminates in a project kickoff meeting, where the team, stakeholders, and other relevant parties are brought together to lay out the project goals, schedule, processes, and the chain of communication.

Step 2: Planning (Generating Concepts)

- The project plan must include every component associated with the execution of the project, including the costs, risks, resources and timeline.
- The project scope (the work required to complete the project) is defined using a work breakdown structure (WBS).
- Project managers often lay out their project plan using a Gantt chart, which provides a visual representation of the entire project. This provides a roadmap for the work until the project reaches its conclusion

Step 3: Executing (Develop A Solution)

- The tasks and milestones outlined in the plan are tackled to produce the deliverable to the client's or stakeholder's satisfaction.
- Along the way, the project manager will reallocate resources as needed to keep the team working. In addition, they will identify and mitigate risks, deal with problems and incorporate any changes.

Step 4: Monitoring And Controlling (Construct and Test Prototype)

- Monitoring the progress and performance
- Quality control procedures are applied to guarantee quality assurance.
- The biggest issues in a project are typically related to three factors—time, cost and scope, which collectively are referred to as the triple constraint.

Step 5: Closing (Evaluate And Present Solution)

- The fifth project management phase is project closure, in which the final deliverables are presented to the client or stakeholder.
- At this point the project manager and team can conduct a post-mortem to evaluate the lessons learned from the project.
- Depending on the project, the closure phase may include handing over control to a different team, such as an operations management team.
- In this case, it is the job of the project manager to ensure such a transition occurs smoothly.

Product and Project Life Cycle

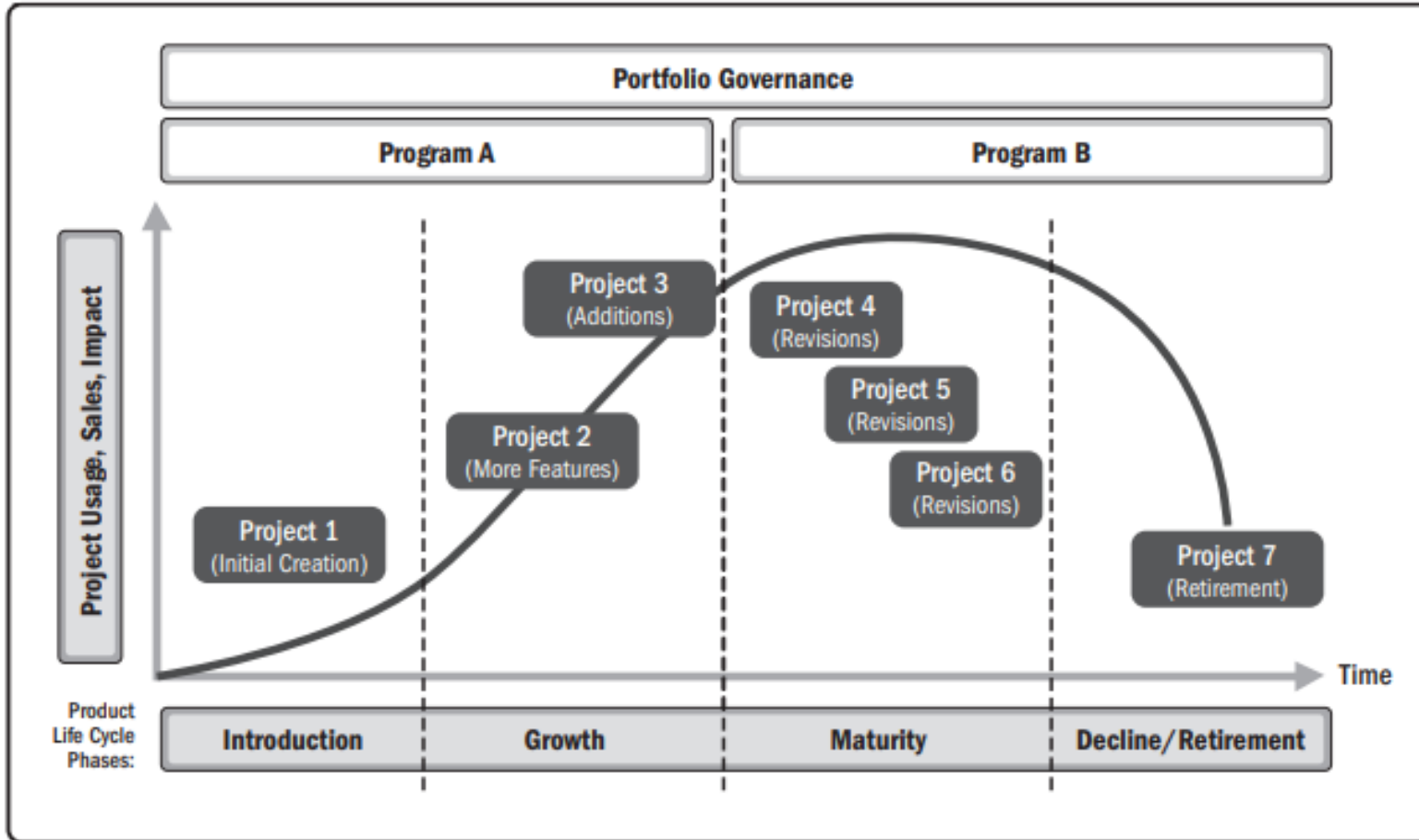


Figure 2-4. Sample Product Life Cycle

Quiz: Are these Projects?

- Building a deck (Yes/No)
- Implementing a new system (Yes/No)
- Mowing the lawn (Yes/No)
- Planning a wedding (Yes/No)
- Planning a fundraiser (Yes/No)
- Planning a student graduation (Yes/No)

Project Organization

- Project organization refers to the style of coordination, communication, and management the project team uses throughout a project lifecycle.
- **Objective:** Create an environment that encourages the development of interactions between team members with very less amount of disruptions, overlaps, and conflicts.



“Project organization is all about thinking ahead.”

Responsibility of Project Organization

- Definition of process even maintenance of project process
- Process automation
- The organization's role or role of process automation is taken and achieved by a single individual or various other teams.

Types of Project Organizational Structures

- **Functional** is when the organizational departments are grouped by areas of specialization. In this case, the project is usually executed in a silo environment.
- **Projectized** is when the entire organization is organized by the project.
- **Matrix** has teams report to both a functional manager and project manager, sort of a hybrid of the previous two structures.
- **Organic** project organization embraces flexibility.
- **Virtual** is when the project manager is the hub in the network.
- **Multi-division** means that functional groups are decentralized.

Six Considerations Can Help Lend Project Success to Any Organizational Setup

1. Clear project requirements
2. Alignment among stakeholders
3. Resources available and needed
4. Communication planning
5. Review and feedback process
6. What could go wrong and possible corrections

Case Study: The Enterprise Resource Planning Project



Adobe Acrobat
Document

1. Which of the following is not a project management goal?

A. Keeping overall costs within budget

B. Delivering the software to the customer at the agreed time

C. Maintaining a happy and well-functioning development team

D. All of the above

2. The process each manager follows during the life cycle of a project is known as:

A. Project Management

B. Manager life cycle

C. Project Management Life Cycle

D. All of the mentioned

3. Quality planning is the process of developing a quality plan applicable for.

A.Team

B. Project

C.Customers

D.project manager

4. Which of the following is an incorrect activity for the configuration management of a software system?

A. Internship management

B. Change management

C. Version management

D. System management

Assignment Questions

- Fill in the blanks:

1. The successful is all about structure, control, sufficient attention to detail and continuously driving action.
2. A project can only be successful when there is no conflict between the and the local populace.
3. Project Management is quite often the province and responsibility of an individual
4. The project manager needs to be an expert who can motivate the workforce by training and promoting leadership among them.
5. Every project has some elements that are

Assignment Questions

- Discuss Project Management as a profession.
 - What do you know about Project Life Cycle? Support your answer with a suitable example.
-
- Points to be kept in mind:
 - Marks for fill ups (0.5 each)
 - Question 1 carry 2.5 marks
 - question 2 carries 5 marks
 - Word limit for 1 max.250 words limit
 - 2 question max. 500 words limit

Old Question Papers

Printed Page 1 of 2

Sub Code: NMBA034-MBTM034

Paper Id:

270324

Roll No:

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MBA/MBA-TM
(SEM-III) THEORY EXAMINATION 2019-20
PROJECT MANAGEMENT

Time: 3 Hours

Total Marks: 100

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. **Attempt *all* questions in brief.** **2 x 10 = 20**

- a. Why feasibility report is prepared before commencement of a project
- b. How would you defend the statement that Project manager must help themselves?
- c. Compare and contrast PERT & CPM
- d. What are the sources of project finance?
- e. Differentiate between the UNIDO approach and LM Approach.
- f. Write external and internal factors of SWOT analysis by giving example
- g. To what extent is forecasting scientific, in comparison to prediction?
- h. Payback period method ignores the time value of money', illustrate with an example.
- i. Write the general principles of shadow pricing?
- j. How technical analysis can be beneficial for project analysis.

Old Question Papers

SECTION B

2. Attempt any *three* of the following:

10x3=30

- a. What are the components of detailed feasibility Study? Explain.
- b. Explain the structure of different types of project organization
- c. Below are given the figures of production (in thousand tones) of a fertilizer factory:

Year	2008	2009	2010	2011	2012	2013	2014
Production	21	21	30	29	62	68	91

Fit a straight line trend by the method of least squares and estimate the trend value for year 2015.

- d. What is technical analysis? Discuss the various factors looked into for doing technical analysis.
- e. Explain various tools and techniques used for Social Cost benefit analysis of a project. How human aspects effects project management.

Old Question Papers

SECTION C

3. Attempt any *one* part of the following: 10x1=10

- a. Select a project in your neighborhood and discuss its objectives, scope and strategies?
- b. Explain the different method of forecasting?

4. Attempt any *one* part of the following: 10x1=10

- a. a. Under a promotional employment plan, it is proposed to plan the sale of news papers on buses during off peak hours. The vendor can purchase the papers at a special concessional price of 30 paisa and sale of it is at 50 paisa a piece. Any unsold copy is a dead loss. A vendor has estimated the following probabilities for the number of copies demanded: CO2

No of copies: 15 16 17 18 19 20

Probability: .4 .1 .20 .20 .05 .05

Prepare payoff table and find out how many copies should be ordered so that the expected profits will be a maximum.

Old Question Papers

Printed Page 2 of 2

Sub Code: NMBA034-MBTM034

Paper Id:

Roll No:

- b. “Organizing systems and procedures for implementing a project is essential for achieving the project objectives”. Support this statement with your arguments in favor or against.

5. Attempt any *one* part of the following: 10x1=10

- a. “Finance is the life blood of any project”. Discuss the statement and Explain the long term sources through which it can be arranged.
- b. How the profitability of project is calculated?

6. Attempt any *one* part of the following: 10x1=10

- a. A project requires a cash outlay of Rs 20000 and generates the cash flow of Rs 8000, 7000, 4000, what is the project pay back?
- b. What is profitability index? How it is different from NPV?

7. Attempt any *one* part of the following: 10x1=10

- a. What is team development? How a good team can be developed?
- b. What are the measurements used for controlling the project?

Expected Questions

- Q1. Discuss the project. elaborate are the components of the project management process.
- Q2. Describe various Tools and techniques of project management.
- Q3. Discuss the characteristics of a Project leader.
- Q4. Explain the term 'project organization'.
- Q5. Explain the project life cycle with a diagram.

Video Links

- https://www.youtube.com/watch?v=39_O-AnzTQg
- <https://www.youtube.com/watch?v=BOU1YP5NZVA>
- <https://www.youtube.com/watch?v=ZKOL-rZ79gs>
- https://www.youtube.com/watch?v=RjOA7AxOVj8&list=PLPjSqITyvDeX77O9g4E_lgJbAmLK5SFu6
- <https://www.youtube.com/watch?v=xzkgMVDReHi4&t=1s>
- https://www.youtube.com/watch?v=1Fqe_7rn3Vs
- <https://www.simplilearn.com/what-is-a-deliverable-article>
- <https://www.projectmanager.com/training/write-scope-work>
- <https://plan.io/blog/scope-of-work/>
- <https://www.projectmanager.com/blog/milestones-project-management>
- <https://www.teamgantt.com/blog/the-how-and-why-of-using-milestones-in-your-project-plan>
- <https://www.projectsmart.co.uk/top-10-qualities-project-manager.php>
- <https://pmtips.net/article/10-characteristics-of-successful-project-teams>
- <https://www.sinnaps.com/en/project-management-blog/importance-of-project-management>

Books:

- Larsen, E.W., Gray C.F., & Joshi, R. (2021). *Project management: The Managerial process*. McGraw Hill.
- Chandra, P. (2019). *Projects: Planning, Analysis, Selection, Financing, Implementation and Review*. McGraw Hill.

REFERENCE BOOK:

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Recap of the unit

- A project is temporary- It has a defined beginning and end in time and therefore defined scope and resources.
- Project Life Cycle is a series of activities which are necessary to fulfill goals or objectives.
- The application of knowledge, skills, tools, and techniques to project activities to meet project requirements. Project management refers to guiding the project work to deliver the intended outcomes.